

MAIN FACP BATTERY CALCULATION

PROJECT	Samana IVY Garden-2 (1B+G+4P+35+Roof) Residential Building @ PLOT NO: 648-8523, Wadi Al Safa 5, DLRC, Dubai, UAE		
CLIENT	M/s. Samana IFM Holding Ltd	MANUFACTURER	M/s. EDWARDS System Technology
CONSULTANT	M/s. Al Hilal Engineering Consultant	SUBMITTAL	FIRE ALARM, VOICE EVACUATION & FIRE TELEPHONE SYSTEM
MEP CONTRACTOR	M/s. ITALTECH CONTRACTING L.L.C	DATE	10.08.2026
FAS SUPPLIER	M/s. AL ARABIA FOR SAFETY & SECURITY LLC		

Connected equipment

Part no.	Description	Qty	Standby / unit	Alarm / unit	Standby total	Alarm total
4-CPU	Central Processor Unit Module	1	211	211	211	211
4-PPS	Primary Power Supply	1	45	45	45	45
3-SDDC2	Dual Loop SIGA Data Controller	5	264	336	1320	1680
4-LCDLE	Liquid Crystal Module	1	40	93	40	93
4-NET-TP	SFP Network controller	2	5	5	10	10
4-FWAL4	Network Firewalls	1	130	130	130	130
3-ZA20A	Zoned Audio Amplifiers	2	62	1120	124	2240
4-AUDTELS	Audio-Telephone Interface Module	1	101	85	101	85
4-MIC	Paging Microphone	1	8	38	8	38
4-FT	Firefighter Master Handset	1	9	168	9	168

Battery sizing

Standby current	1,998.000 mA
Alarm current	4,700.000 mA
Standby, 24 h	47.952 Ah
Alarm, 30 min	2.350 Ah
Base requirement	50.302 Ah
With 20% spare	60.362 Ah

SELECTED BATTERY2 NOS 65AH

Calculation notes

- 1- The calculation has been carried out based on the full-load condition. The current consumption of 3-SDDC2 Dual Loop Cards has been considered based on a maximum of 500 connected devices, including all detectors and modules.
- 2- A separate cabinet is provided for the NAC circuits, utilizing SIGA-BPS10A/230. Refer to the NAC Circuit Calculation and SIGA-BPS10A/230 Battery Calculation for details.
- 3- A separate cabinet is provided for the VES circuits, utilizing SIGA-APS6A/230. Refer to the VES Circuit Calculation and SIGA-APS6A/230 Battery Calculation for details.
- 4- 24 hours of standby operation and 30 minutes of alarm operation have been considered in the battery calculations, in accordance with DCD requirements and NFPA 72 standards.